

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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Anonymous submission

ABSTRACT

Reliable evaluation of generative models requires preference signals that automatic metrics often approximate only indirectly. Human judgments provide this signal, but they are expensive to collect and statistically noisy because raters vary in expertise, attention, and diligence. Pairwise comparisons reduce some response variability, yet converting sparse and heterogeneous comparisons into global quality estimates remains a statistical aggregation problem rather than a counting problem. Bradley-Terry aggregation is a natural foundation, but standard Bayesian variants typically treat raters as exchangeable sources of evidence, whereas existing rater-reliability extensions often rely on gradient-based optimization and provide limited uncertainty interpretation. We introduce BBQ, a Bayesian Bradley-Terry model for maximum-a-posteriori aggregation that jointly estimates item quality and rater-specific reliability under a simple mixture model of informed preferences and random guesses. The resulting latent-variable formulation yields closed-form Expectation-Maximization updates with monotonic improvement of the observed-data objective and produces interpretable item- and rater-level parameters. Across human-preference datasets for language and image-compression systems, including a newly collected dataset with screened and unscreened raters, BBQ improves ranking stability and top-item recovery most clearly in noisy or heterogeneous-rater settings while remaining computationally efficient on cleaner benchmarks where the gains over simpler Bradley-Terry variants are smaller. BBQ therefore provides a practical and transparent route from noisy pairwise preferences to reproducible human-evaluation rankings.

1 INTRODUCTION

The evaluation of generative models increasingly depends on human preference data, particularly for large language models (LLMs) and image-generation systems whose perceived quality is only partially captured by automatic metrics. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, remain useful diagnostic quantities but provide incomplete substitutes for human judgment (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). As a result, human evaluation continues to anchor meaningful comparative rankings. Leaderboards and empirical studies for both large language models and Learned Image Compression (LIC) methods consequently rely on preference evidence rather than automatic scores alone.

The central practical difficulty is that human evaluation is costly, time-consuming, and sensitive to subjective variation; an evaluation protocol must therefore extract reliable information from limited and noisy judgments. Pairwise comparison protocols address part of this difficulty by asking participants to choose between two items rather than assign absolute scores, a format that is easier for raters and often yields more consistent responses (Zerman et al., 2018; Wang et al., 2023). Exhaustive pairwise evaluation remains infeasible, however, because the number of candidate pairs grows quadratically with the number of items. In sparse comparison designs, naive win rates are inadequate global quality scores: the evidential value of a win depends jointly on the opponent’s quality and on the reliability of the rater who supplied the judgment.

Transforming sparse pairwise outcomes into leaderboard-ready scores therefore requires an aggregation model that is statistically principled, computationally efficient, and explicit about the reliability

of its evidence. The Bradley-Terry (BT) model (Bradley & Terry, 1952) supplies the standard probabilistic formulation by inferring latent item scores from observed comparison outcomes. Because it converts local preferences into globally comparable skill parameters, BT aggregation has become a central tool for pairwise-judgment analysis across domains.

The classical BT model nevertheless leaves two issues unresolved for realistic human-evaluation studies. First, BT parameters may be estimated by iterative scaling methods (Ford Jr, 1957), gradient descent, or minorization-maximization (MM) algorithms (Hunter, 2004), but these procedures differ substantially in their guarantees. MM reaches the global maximum for the basic BT model, whereas gradient-based procedures, including Elo-style updates widely used in practice, do not provide the same guarantee (Hunter, 2004). For BT extensions that include factors such as home-field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), even MM algorithms may converge only to local optima (Hunter, 2004). Second, human evaluations are heterogeneous: participants may rush, guess, lose focus, or differ in expertise, producing raters whose comparisons have systematically different reliability. When this variability is left unmodeled, noisy raters can distort item-score estimates and reduce the reproducibility of the induced ranking.

We address these limitations with BBQ, a Bayesian Bradley-Terry model for MAP aggregation that jointly estimates item quality and rater reliability. BBQ occupies a specific position within the broader family of annotator-reliability and BT extensions: it retains the interpretability of BT item scores, adds an explicit rater-quality parameter, and derives closed-form EM updates for the resulting Bayesian objective. Where prior noisy-rater BT approaches commonly use gradient-based posterior optimization, BBQ provides monotonic improvement of the observed-data objective and yields directly interpretable item- and rater-level parameters. The Bayesian formulation regularizes skill estimates and supports posterior uncertainty approximations, while the rater-specific quality parameters determine how strongly each participant’s comparisons influence the inferred ranking. Through a latent-variable construction, we derive an EM algorithm that efficiently updates both item skills and rater qualities. The model consequently preserves information from partially reliable raters while attenuating the contribution of inattentive or inconsistent participants. Because the uncertainty approximation is tied to the same ranking model, BBQ also supports more interpretable comparisons across datasets and studies.

We evaluate BBQ on several human-evaluation datasets and introduce a newly collected *inhomogeneous rater quality (IHQ)* dataset¹. The experiments show rapid convergence, strong computational efficiency, and improved stability in the noisy and mixed-quality settings for which rater modeling is most consequential; on cleaner datasets with homogeneous raters, BBQ remains competitive and the margins over simpler BT variants are correspondingly smaller. The evidence therefore supports BBQ as a practical aggregation framework for studies in which noisy pairwise comparisons must be converted into reliable global rankings, with particular value in crowdsourced or otherwise heterogeneous evaluation pipelines. Taken together, the model, derivation, and empirical analysis provide a cost-conscious route toward more interpretable and reproducible human studies of AI-generated content.

2 RELATED WORK

Ranking items from human judgments is a recurring requirement in empirical machine learning, where evaluation often depends on preferences that cannot be fully reduced to automatic metrics. Pairwise protocols, in which participants choose between two alternatives rather than assign a score to a single item, increase sensitivity to subtle perceptual differences and reduce response variability (Zerman et al., 2018; Wang et al., 2023). Many model leaderboards therefore use pairwise comparisons as their primary source of human-preference evidence. The rater supplies local binary preferences, and an aggregation model converts those local observations into a global ranking. Prominent examples include Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), both of which collect pairwise comparisons and aggregate them using variants of the Bradley-Terry (BT) model.

¹The IHQ dataset will be released upon acceptance.

The Bradley-Terry model (Bradley & Terry, 1952), originally introduced for ranking sports competitors, specifies a probabilistic relationship between latent item skill and the probability that one item is preferred to another. By estimating item scores from observed wins and losses, it provides the canonical mechanism for transforming pairwise comparison data into a global ordering. Its influence extends across games, consumer-preference analysis, and machine-learning evaluation, and a substantial literature has generalized the model to richer comparison structures. The Plackett-Luce model, for example, extends the BT framework from pairwise comparisons to rankings over multiple items (Plackett, 1975; Luce et al., 1959) by defining a probability distribution over permutations through successive BT-style choices (Luce et al., 1959). Other extensions incorporate context-specific factors, including home-field advantage (Agresti, 2010), ties (Rao & Kupper, 1967), and comparisons between groups rather than individuals (Huang et al., 2006).

Parameter estimation for the basic BT model is commonly posed as maximum likelihood estimation (MLE). Zermelo (1929) introduced an iterative procedure for computing these estimates, and that procedure remains foundational. Lange et al. (2000) later showed that the procedure is a special case of minorization-maximization (MM), in which each iteration optimizes a tractable surrogate objective. Hunter (2004) extended MM algorithms to generalized BT models and identified conditions under which they converge to the MLE. For the classical BT model, convexity ensures convergence to the global optimum (Hunter, 2004). For extensions involving home-field advantage, multiple comparisons, or ties, however, MM algorithms may only reach a local maximum (Hunter, 2004).

Bayesian BT formulations incorporate prior information, regularize skill estimates, and produce uncertainty estimates rather than point rankings alone (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) established that MM algorithms can be interpreted as Expectation-Maximization (EM) procedures, enabling Bayesian inference via Gibbs sampling. This latent-variable formulation yields tractable complete-data likelihoods and supports convergence of the resulting Markov Chain Monte Carlo methods. The Bayesian perspective smooths posterior estimates, reduces sensitivity to poorly conditioned data, and quantifies uncertainty in inferred item skills.

Annotator-quality models address a complementary limitation of the standard BT assumption: in human studies, comparisons are not equally reliable. This issue is especially pronounced in crowd-sourced evaluations, where participants may differ substantially in expertise and care, motivating models that explicitly represent annotator reliability (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters. Their formulation uses Gaussian priors for item and rater parameters and likelihood scaling factors that modulate each annotator’s influence. Posterior inference is then performed with gradient-based optimization, which can be sensitive to initialization and local optima in high-dimensional parameter spaces. Our EM-based formulation targets the same practical problem of heterogeneous rater quality while providing monotonic improvement of the observed-data objective. The Elo-based rater model developed by Google Research implements a simplified version of this line of work and is used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and related studies (Mentzer et al., 2020; Ballé et al., 2025).

BBQ is positioned at the intersection of Bayesian BT estimation and annotator-reliability modeling. Rather than claiming a new class of reliability models, we contribute a BT aggregation framework that combines a Bayesian rater-quality parameterization with closed-form EM updates. In contrast to Chen et al. (2013), who combine Gaussian priors with gradient-based posterior optimization, BBQ uses conjugate priors and an EM algorithm that monotonically improves the observed-data objective. In contrast to the Elo-based rater model used in CLIC (CLIC, 2025), which relies on simplified Elo-style updates, BBQ supplies a Bayesian MAP treatment with interpretable rater-quality parameters and posterior uncertainty approximations for item skills.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

We consider the problem of converting noisy pairwise comparisons into a reliable ranking over a finite item set. Let K denote the number of items and let R denote the number of raters who compare these items in pairs. The observed dataset, denoted by D , consists of the resulting binary

comparison outcomes. For two items i and j , the Bradley-Terry model (Bradley & Terry, 1952) assigns the preference probability

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ is the latent skill parameter associated with item $k \in \{1, 2, \dots, K\}$. The model provides an interpretable mechanism for inferring item scores from observed wins and losses. Its basic form, however, assumes that all comparisons are equally informative. Human-evaluation data often violate this assumption because participants differ in expertise, attention, and diligence, and these differences induce comparison-level noise that is correlated with rater identity.

To account for heterogeneous reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. The model treats each comparison as arising from a two-component mixture: with probability q_r , rater r produces an informed judgment governed by the Bradley-Terry preference probability; with probability $1 - q_r$, the rater produces an uninformative random response, modeled as a fair coin flip between the two items. Thus the probability that rater r ranks item i above item j is

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The observed-data log-likelihood is

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ denotes the total number of comparisons between items i and j by rater r , and $w_{r,ij}$ is the number of times rater r ranks item i above item j . This likelihood does not admit a closed-form maximum-likelihood estimator for λ and q . Gradient-based optimization is possible, but it introduces learning-rate choices and may be sensitive to initialization. We instead use Expectation-Maximization (EM), which avoids learning-rate tuning and guarantees monotonic improvement of the observed-data likelihood with convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we use a Thurstonian interpretation of the Bradley-Terry model (Diaconis, 1988). A comparison between items i and j is represented as a race in which the items have random arrival times Y_i and Y_j . These arrival times follow exponential distributions:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller arrival time is declared the winner. This construction recovers the standard Bradley-Terry probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we introduce latent variables that make the complete-data likelihood tractable. First, define an indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which indicates whether the c -th comparison of items i and j by rater r is generated by the Bradley-Terry component rather than the random-guessing component. Using these indicators, define $Z_{r,ij}$ as the sum of the minimal arrival times across the $n_{r,ij}$ comparisons by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditioned on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the variable $Z_{r,ij}$ follows a Gamma distribution,

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma latent-variable representation is the mechanism that yields tractable EM updates while retaining rater-specific quality parameters.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm computes maximum a posteriori (MAP) point estimates of the model parameters λ and q by optimizing an expected complete-data log-posterior. The latent quantities are the mixture indicators and the unobserved arrival-time variables induced by the Thurstonian construction. The resulting procedure is guaranteed to converge to a stationary point of the posterior objective.

We place a Gamma prior $\lambda_k \sim \Gamma(a, b)$ on each item skill, exploiting conjugacy with the exponential arrival-time representation, and a Beta prior $q_r \sim B(\alpha, \beta)$ on each rater-quality parameter.

At each iteration, EM maximizes the expected complete-data log-posterior conditioned on the current parameter estimates (λ^*, q^*) . The algorithm alternates between the following two steps until convergence:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, denoted by Q . This quantity is the conditional expectation of the complete-data log-posterior under the observed data and the current parameter estimates:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A,Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood ℓ_c decomposes into three terms:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the model parameters by maximizing Q . Using the conditional expectations computed in the E-step, this maximization reduces to closed-form updates. The central quantity in both updates is the posterior probability that a comparison from rater r was generated by the Bradley-Terry component. We denote this responsibility by γ ; it weights each judgment according to its consistency with the current model estimates:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability that item i beats item j . Thus γ is the current posterior responsibility assigned to the informed-comparison component; larger values indicate comparisons that the model treats as more informative.

The update for q_r is a prior-regularized weighted average of the comparisons attributed to the informed component. The numerator accumulates the rater's effective non-random comparisons, weighted by γ , and adds the contribution of the Beta prior; the denominator normalizes by the rater's total comparison count and prior mass. The update is

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The update for λ_i balances effective wins against exposure to competing items. The numerator aggregates the γ -weighted wins of item i across raters and adds the Gamma-prior contribution, while the denominator normalizes by the expected comparison exposure against all other items, again including the Gamma-prior rate term. The update is

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

The full derivation of the Expectation-Maximization updates is provided in Appendix B.

4 EXPERIMENTAL RESULTS

We evaluate the **Bayesian Bradley-Terry** model with rater **Quality (BBQ)** against two BT-based baselines: Bayesian Bradley-Terry without rater-quality parameters (Bayes-BT) (Caron & Doucet, 2012) and a gradient-descent BT model with rater quality (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025), which was also used by CLIC (2025) and Ballé et al. (2025). The experimental hyperparameters are reported in Appendix C.

The evaluation spans human-preference benchmarks for large language models (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image compression (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To isolate the role of rater quality, we partition IHQ into screened and unscreened subsets. This design allows us to compare model behavior under high-quality, low-quality, and mixed-quality comparison regimes. Details of IHQ collection are given in Appendix E, and a summary of all datasets, including comparisons, raters, and items, is provided in Appendix D and Table 2.

Because independent ground-truth rankings are unavailable for the real datasets, our primary empirical target is ranking stability relative to a full-data estimate rather than absolute correctness against an external oracle. We first validated the models on simulated datasets, where all methods recover the same ordering as the number of samples increases. For the crowdsourced datasets, we use the ranking inferred from the complete dataset as an empirical reference and interpret bootstrap agreement as a measure of reproducibility under resampling. This target is supported by the top-1 behavior of the real datasets: for all datasets except IHQ-unscreened, all three models recover the same top-ranked item from the full data. On IHQ-unscreened, Crowd-BT and Bayes-BT do not identify the reference image as the highest-quality item, whereas BBQ does.

A reliable aggregation method should produce similar rankings when the study is resampled or repeated under comparable conditions. We approximate this reproducibility using bootstrapping, which estimates variability in the induced rankings. Bootstrapping details are provided in Appendix F. The WD and HiFiC studies used active selection of comparison pairs: HiFiC used a binary-search strategy, and WD selected pairs by maximum information gain. Consequently, bootstrap results on these two datasets should be interpreted as stability diagnostics rather than as exact replications of the original sampling process.

We report two complementary stability metrics. Top-1 agreement is the fraction of bootstrap samples that identify the same best item as the full-data ranking. Kendall’s Tau (τ) measures ordinal correlation between the bootstrap ranking and the full-data ranking (Kendall, 1938), with higher values indicating greater stability. Top-1 agreement captures reproducibility of the winner, whereas Kendall’s Tau summarizes stability of the overall ordering.

4.1 PERFORMANCE ACROSS DATASETS

We compare the three aggregation methods across datasets to assess ranking stability under different rater and comparison regimes. Table 1 reports Top-1 agreement and Kendall’s Tau for each dataset and method. Datasets with more comparisons per item, such as MT-Bench, WD, and HiFiC, are generally easier, although HUMAINE departs from this pattern and indicates that comparison count alone does not determine stability. Rater consistency, rater diversity, and sampling design also affect the reliability of the inferred ordering.

Across the eight evaluated settings, BBQ provides the strongest aggregate stability profile. It identifies the top-performing item most frequently, including shared-best performance on three datasets. For Kendall’s Tau, BBQ is best on five datasets and second-best on the remaining three, indicating that the rater-quality component improves overall ranking stability without degrading performance on cleaner settings.

The three datasets on which BBQ ranks second in Kendall’s Tau are comparatively clean or homogeneous. On MT-Bench, all models perfectly recover the top item. The WD dataset, collected by Ballé et al. (2025), includes only five raters, likely the paper’s authors, and therefore reflects an unusually

Table 1: Bootstrap stability of BT-based aggregation methods across eight datasets. Top: Top-1 agreement [%] with the full-data ranking; Bottom: Kendall’s Tau between bootstrap and full-data rankings. BBQ most frequently recovers the full-data top item, achieves the highest Kendall’s Tau on five datasets, and ranks second on the remaining three.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	<u>0.9245</u>	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

careful evaluation setting. In IHQ-screened, raters were explicitly filtered for quality. These results clarify the scope of the benefit: when rater quality is uniformly high, explicitly modeling rater reliability offers limited additional information, and the margins over simpler BT variants naturally shrink.

The IHQ results make the role of rater modeling especially clear. BBQ substantially outperforms Crowd-BT and Bayes-BT when low-quality raters are present, as in IHQ-all and IHQ-unscreened. All three methods achieve their highest Top-1 agreement on IHQ-screened, confirming that rater selection remains valuable. However, Crowd-BT drops markedly on the full dataset despite explicitly modeling rater quality, plausibly because crowdsourced raters provide fewer than 40 comparisons each and the per-rater reliability estimates are therefore difficult to stabilize. BBQ maintains strong Top-1 agreement without screening, with only a small decrease relative to IHQ-screened, which is consistent with the intended effect of its rater-quality mixture model.

Uniformly high rater quality is difficult to guarantee in practice. Large-scale crowdsourcing introduces variability, screening procedures increase cost, and subjective factors can affect even diligent annotators. In this setting, BBQ offers a principled way to retain information from partially reliable raters while reducing the impact of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As observed in Table 1, datasets with more comparisons per model generally yield better performance. The number of comparisons can be increased in two ways: by adding more raters, or by increasing the number of comparisons each rater performs.

Figure 1 illustrates the impact of both factors on the three BT variants. The left column shows performance as a function of the number of raters (with the number of comparisons per rater fixed at the maximum), while the right column shows performance as a function of the number of comparisons per rater (with the number of raters fixed at the maximum). A clear difference in performance can be observed in Top-1 accuracy, while Kendall’s τ remains similar across models. Crowd-BT fails to converge with only one or two raters, highlighting the advantage of using the EM algorithm, which provides convergence guarantees compared to gradient descent.

Crowd-BT and BBQ perform similarly when the number of raters or comparisons per rater is small. BBQ requires both multiple raters and multiple comparisons per rater to effectively distinguish between rater qualities. As the number of raters or comparisons per rater increases, BBQ increasingly outperforms Bayes-BT. In contrast, Bayes-BT tends to underperform overall but can surpass Crowd-BT when limited data per rater or a few number of raters are available, since Crowd-BT cannot reliably estimate rater quality in such sparse settings.

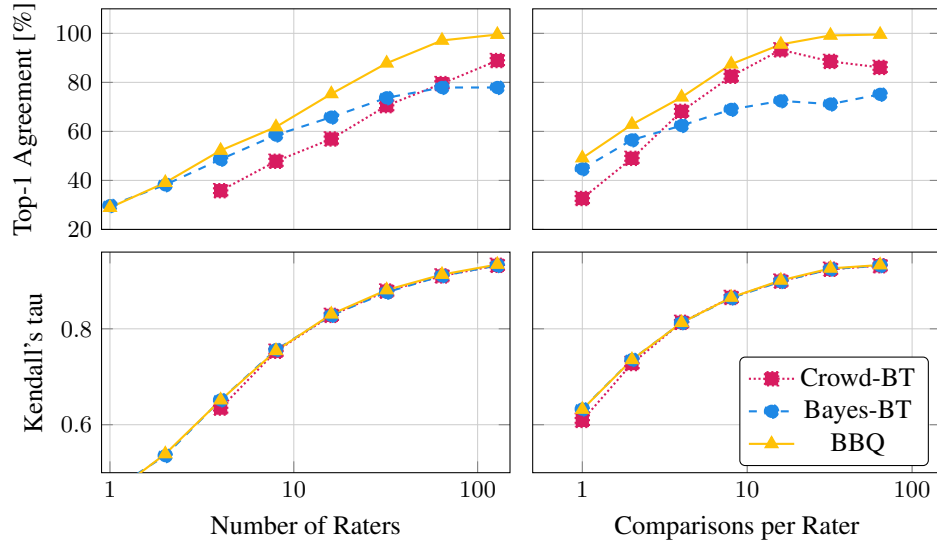


Figure 1: Scaling behavior of Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** Performance vs. number of raters. **Right:** Performance vs. number of comparisons per rater. Both Top-1 agreement and Kendall’s τ improve noticeably with more raters or comparisons. While Top-1 agreement differentiates between models, Kendall’s τ remains similar across models. Crowd-BT fails to converge with very few raters, highlighting the EM algorithm’s advantage. Crowd-BT and BBQ perform similarly under sparse data, but BBQ outperforms Bayes-BT as the number of raters or comparisons grows.

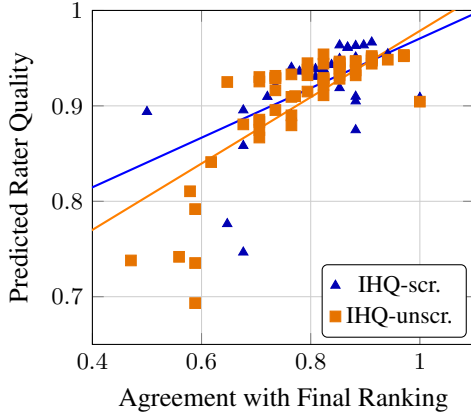


Figure 2: Scatter plot of rater agreement with the final ranking (x-axis) versus the predicted rater quality (y-axis) for the IHQ datasets. Each point corresponds to an individual rater. Triangles denote the filtered dataset, and squares denote the unfiltered dataset.

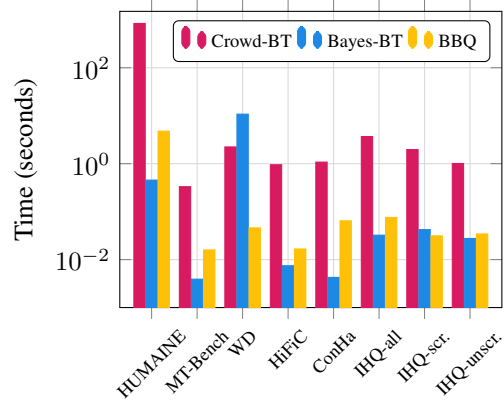


Figure 3: Average computation time in seconds (log-scale) for three models measured on a single bootstrapped sample across eight datasets. While Crowd-BT can require substantial computation time on some datasets, BBQ consistently remains fast across all datasets.

4.3 RATER QUALITY

Figure 2 shows the relationship between agreement with the final ranking and predicted rater quality. We observe a clear positive correlation: raters who agree more closely with the consensus ranking are assigned higher quality by the model. For the filtered dataset, the Pearson correlation is $r = 0.551$ across 50 raters. For the unfiltered dataset, the correlation is stronger, with $r = 0.724$ across 62 raters.

As expected, the unscreened dataset contains several raters with both lower agreement and lower predicted quality. Some raters in the IHQ-unscreened dataset even fall below the level of random guessing (50% agreement), systematically disagreeing with the majority. This highlights the importance of modeling rater quality when aggregating pairwise comparison data. BBQ successfully identifies such raters and assigns them lower quality scores, thereby reducing their influence on the final ranking and mitigating the noise they introduce.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average computation time in seconds required by each method to process a single bootstrapped sample across different datasets. These timings provide a practical perspective on the feasibility of the methods in real-world evaluation scenarios. BBQ consistently converges within a few seconds on all datasets. Crowd-BT requires more time than BBQ across the board, with particularly long runtimes on datasets with many comparisons, such as HUMAINE, where convergence takes around 15 minutes. Bayes-BT converges slowly on the WD dataset, though on other datasets it is slightly faster than BBQ. The efficiency of BBQ stems from the closed-form EM updates derived in our method, which enable rapid convergence even on large datasets.

It is also important to note that Crowd-BT was highly optimized and implemented in C (Kernighan & Ritchie, 1988), whereas BBQ was implemented using plain NumPy without specific optimizations. This suggests that BBQ could be made even faster with a compiled or vectorized implementation. Consequently, BBQ is not only more robust and stable but also highly practical for large-scale human preference studies or applications that require repeated bootstrapping. The combination of accuracy, stability, and speed makes BBQ a compelling choice for real-world deployments.

4.5 CONFIDENCE INTERVALS

We evaluate how Crowd-BT, Bayes-BT, and BBQ estimate uncertainty in item rankings. In a controlled simulation, raters provide pairwise comparisons of equally skilled items (H_0). Each model estimates skills and constructs uncertainty intervals to test for significant differences (Appendix G). The resulting Type I error rates are shown in Figure 6. At the 99% confidence level, the expected error rate is 1%. Crowd-BT and BBQ align closely, yielding slightly lower rates, while Bayes-BT is more conservative at $\sim 0.1\%$. Thus, Crowd-BT and BBQ provide well-calibrated uncertainty estimates, whereas Bayes-BT underestimates false positives.

5 LIMITATIONS

Although BBQ effectively models rater quality, its advantages diminish in settings where all raters are uniformly reliable. In such homogeneous datasets, modeling variability provides little additional benefit. Ensuring consistently high-quality raters, however, often requires substantial cost and effort, which may not be feasible in large-scale studies. This tension highlights that BBQ is most useful in realistic crowdsourced settings, but less so when evaluations are carefully curated.

Another limitation concerns the type of data the framework can handle. BBQ is currently restricted to pairwise comparisons, whereas many human evaluation studies use ratings, rankings, or multi-way inputs. Extending the model to handle these forms of feedback would broaden its applicability.

From a methodological perspective, our experiments rely on a limited number of bootstrapped comparisons. While this provides a practical measure of stability, larger-scale studies would be needed to further validate robustness. Additionally, the model assumes independence across comparisons and does not account for contextual or order effects, which may influence human judgments in practice.

Finally, although BBQ scales well computationally, extremely large numbers of items or raters could still pose challenges without optimized or compiled implementations.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model that jointly estimates item quality and rater reliability. Our core contribution is the derivation of an Expectation-Maximization (EM) algorithm

486 that simultaneously estimates item skills and individual rater quality, effectively down-weighting or
487 removing the influence of unreliable participants. By explicitly modeling rater quality, the method
488 produces more stable and accurate rankings. The EM algorithm ensures rapid convergence and
489 monotonic likelihood improvement, addressing limitations of gradient-based approaches.

490 Across diverse datasets, BBQ consistently achieved high Top-1 agreement and superior Kendall’s
491 Tau, demonstrating both robustness and reliability. The model excels in scenarios with noisy or
492 crowdsourced raters, maintaining accuracy even when raters contribute few comparisons. BBQ is
493 particularly effective in large-scale settings where many raters are non-experts or vary widely in at-
494 tentiveness and diligence. Our results highlight that incorporating rater quality is especially crucial
495 when evaluation quality is heterogeneous or partially unreliable. Additionally, the Bayesian frame-
496 work provides principled uncertainty estimates for item scores, enabling interpretable comparisons
497 across studies. We further demonstrate that the model’s error bars are well-calibrated and can be
498 used to assess whether differences between items are statistically significant. Predicted rater quality
499 aligns with agreement to final rankings, validating the model’s ability to identify reliable evaluators.

500 Overall, BBQ advances human evaluation methodology by offering a practical, interpretable, and
501 generalizable approach to aggregate noisy pairwise comparisons. This work contributes a significant
502 step toward more cost-effective, interpretable, and reproducible human studies for evaluating AI-
503 generated content.

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A USAGE OF LLMs

During the preparation of this paper, we utilized large language models (LLMs) as supportive tools. ChatGPT, Claude, Gemini, and Grammarly assisted with spellchecking, refining phrasing, and condensing text to enhance clarity and readability. Additionally, ChatGPT, Claude, and Cursor were used for code analysis, completion, and the generation of visualizations to aid our development workflow. These tools were employed solely as auxiliary aids, while all core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Each rater r has quality q_r , meaning that with probability q_r they follow the Bradley–Terry model, and with probability $1 - q_r$ they choose randomly. The following is a standard EM (Dempster et al., 1977) derivation that incorporates these rater-specific reliabilities into the Bayesian estimation framework.

B.1 MODEL DEFINITION

Notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley-Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley-Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley-Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley-Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

Bradley-Terry probability:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Mixture with rater quality:

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

Latent exponential view:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

We need to compute:

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

Log-Likelihood of $P(Z \mid A, \lambda)$:

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

Log-Likelihood of $P(A \mid q)$:

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

Log-Likelihood of $P(D \mid \lambda, A)$:

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Complete-Data Log-Likelihood:

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function w.r.t. the parameters (λ, q) , holding the expectations computed in the E-step fixed.

Update for q_r The update for each rater quality q_r is obtained by maximizing Q with respect to q_r (including the Beta prior).

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i (including the Gamma prior).

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the last line gives the explicit update at iteration t .

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with the default hyperparameters.

The only hyperparameters in the Bayes-BT and BBQ models are the prior distribution parameters and the stopping thresholds. For both Bayes-BT and BBQ, we consider the model converged when no ELO score changes by more than 1 between two iterations. The ELO score can be calculated from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where we set the ELO.SCALE.FACTOR to 400.

We chose a gamma prior with shape 5 and rate 0.1 for the skill parameters in both Bayes-BT and BBQ. For BBQ, the beta prior on the rater quality has $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate our models on a diverse set of human preference datasets covering both language and image domains. Table 2 provides an overview.

For natural language, we use the HUMAINE dataset (Team, 2025), a large-scale benchmark with over 100k comparisons, and MT-Bench (Zheng et al., 2023), which provides model comparison judgments from crowd workers on multi-turn dialogues.

For image compression, we consider three datasets: WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. These datasets vary substantially in scale, number of raters, and rater expertise, providing a broad testbed for robustness.

Finally, we introduce the inhomogeneous rater quality (**IHQ**) dataset, obtained from a user study on the CLIC2024 (CLIC, 2025) data conducted via the *Mabyduck* platform. It contains two-alternative-forced-choice (2AFC) judgments across 28 generative image compression models. To study the impact of rater quality, we provide two subsets: (i) **screened**, where raters passed pre-screening checks for attention and display quality, and (ii) **unscreened**, where all raters are included. The screened subset represents higher-quality raters, whereas the unscreened subset better reflects the noisy conditions typical of large-scale human evaluations.

This collection of datasets allows us to study both large-scale, relatively clean benchmarks and smaller, noisier settings where modeling rater reliability is critical.

Table 2: Summary of datasets used in our experiments. The IHQ dataset was collected by us on the Mabyduck platform and includes both 2AFC and 3AFC settings. For this dataset, we provide screened and unscreened subsets: screened subsets include only raters who passed pre-screening tests for attention and display quality, ensuring higher reliability, while unscreened subsets include all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

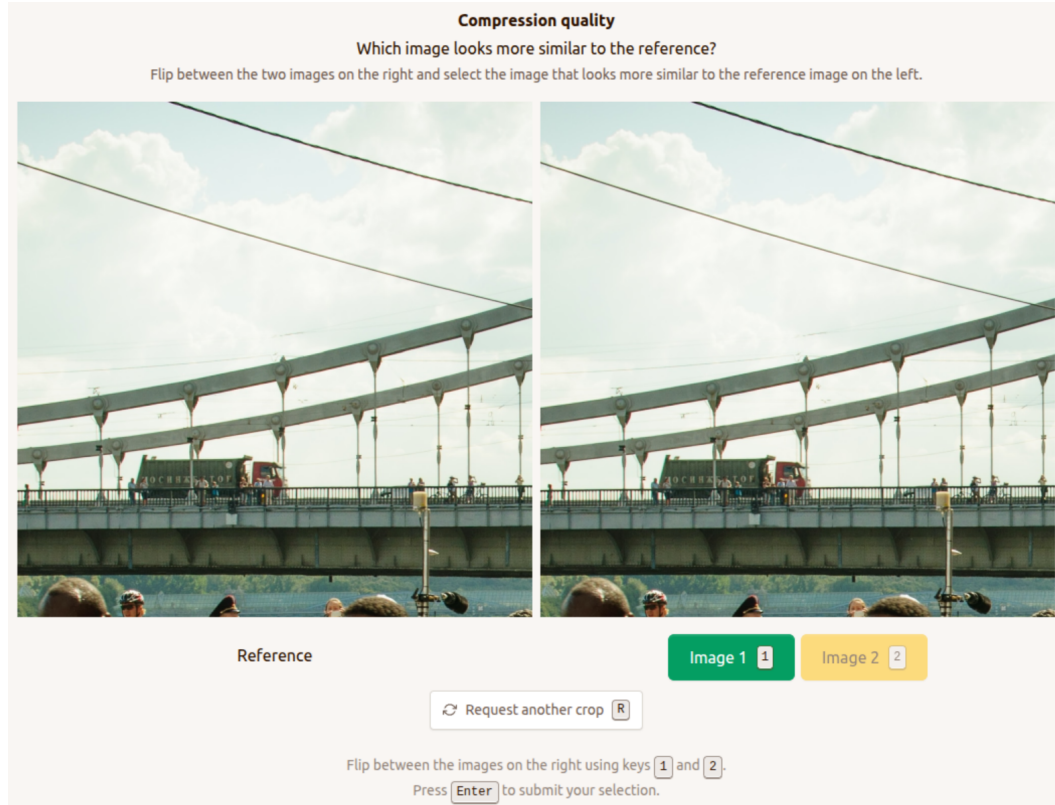


Figure 4: Screenshot of the Mabyduck user study platform used for collecting pairwise comparisons. A reference image is shown on the left, and the rater selects between two compressed images on the right.

E USER STUDY PLATFORM

All pairwise comparisons on the CLIC2024 (CLIC, 2025) dataset were collected using the Mabyduck platform (Ltd., 2025). A screenshot of the platform can be seen in Figure 4. The task asks: “Which image looks more similar to the reference image?” Ties are not allowed, and all pairs were selected uniformly at random.

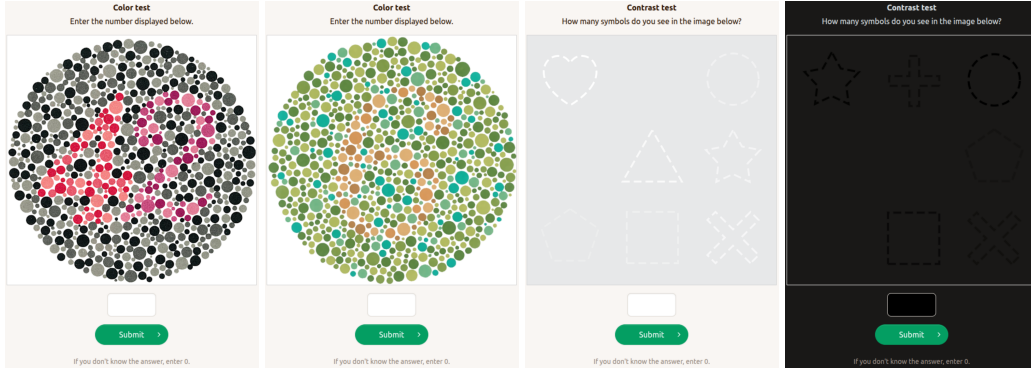


Figure 5: Four example images from the pre-screening process for raters. The first two are color blindness tests, where raters must identify the number displayed in each pattern. The last two are shape detection tests designed to evaluate sensitivity to low-contrast objects: one light gray shape on a white background and one dark gray shape on a black background.

To study rater quality, we split the IHQ dataset into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and completed training comparisons choosing the higher-quality image to reinforce evaluation criteria.

Figure 5 shows four example images from the pre-screening process. The first two are standard color blindness tests, where raters must correctly identify the number shown in each pattern. The final two images are shape detection tests designed to evaluate the raters’ sensitivity to low-contrast objects: one features a light gray shape on a white background, and the other a dark gray shape on a black background. These pre-screening tests help ensure that only raters with adequate visual capabilities contribute to the screened subset.

F BOOTSTRAPPING DETAILS

Since both BBQ and Crowd-BT explicitly model rater quality, we perform bootstrapping over raters rather than individual comparisons. This approach preserves each rater’s comparison distribution and ensures a fair assessment of stability. For all datasets, we perform 10,000 bootstrap resamples of raters, except for the HUMAINE dataset, where a single bootstrap iteration for Crowd-BT takes approximately 15 minutes, as discussed in Section 4.4. On the HUMAINE dataset, we perform 1,000 resamples to reduce computation time.

G UNCERTAINTY ESTIMATION DETAILS

In addition to ranking items, estimating uncertainty is important to assess whether observed differences are statistically significant.

The Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) models quantify uncertainty via the posterior distribution over item skills. Each skill has a Gamma prior, which is updated using the observed pairwise comparisons. Credible intervals derived from the posterior are then converted to the Elo scale to facilitate comparison across items.

The classical Crowd-BT model, being non-Bayesian, estimates uncertainty using a frequentist approximation. Specifically, a second-order Taylor expansion around the maximum likelihood estimate is employed. The Hessian of the log-likelihood is inverted to obtain the covariance matrix, whose diagonal entries correspond to the variances of individual items. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the constants scale the standard deviation to match the 99% confidence level. Narrower likelihood peaks yield smaller intervals, reflecting higher certainty, while flatter peaks produce larger intervals, indicating greater uncertainty.

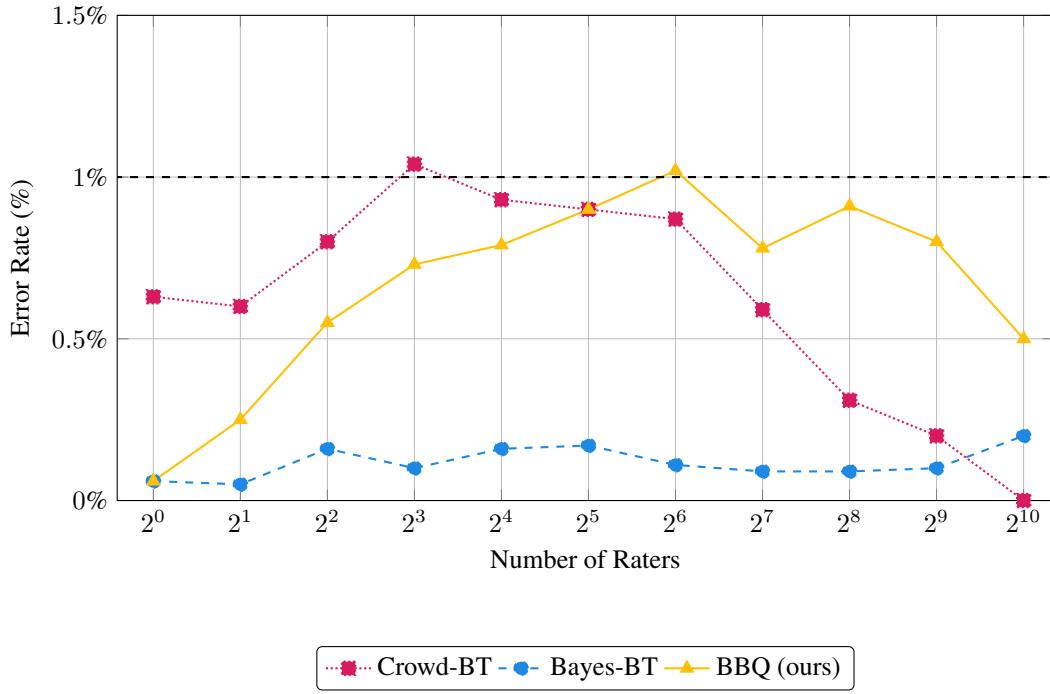


Figure 6: Type I error rate as a function of the number of raters. The dashed line indicates the expected error rate of 1%.

To evaluate how well these models estimate uncertainty, we conduct a controlled simulation experiment. We generate trials with two items of equal skill (50–50 win probability) and simulate pairwise comparison data using coin flips. For each trial, multiple raters (users) are simulated, and models are applied to estimate Elo scores and their associated uncertainty.

Formally, the null hypothesis H_0 states that the two items are equally strong. For each trial, we compute the 99% confidence (or credible) interval for each item’s skill estimate. A model is said to incorrectly reject H_0 if the confidence intervals do not overlap, indicating a statistically significant difference between items when none exists. The primary metric of interest is the frequency with which each model incorrectly concludes that the items are different, i.e., the observed Type I error rate.

We perform 10,000 trials, using 50 comparisons per rater, while varying the number of raters to assess how uncertainty estimates scale with the amount of data. This setup allows us to estimate the empirical Type I error rate for each model and compare it against the theoretical expectation of 1% at the 99% confidence level. As shown in Figure 6, Crowd-BT and BBQ exhibit Type I error rates close to the expected 1%, indicating that their uncertainty estimates are well-calibrated. In contrast, Bayes-BT is overly conservative, consistently producing lower error rates than expected, which suggests its credible intervals are not as well-calibrated for significance testing.